

8.3. Aggregate Planning Using Linear Programming

IE31500-SCM

Demand Forecast at Red Tomato Tools

Sets

$$t \in \{1, 2, 3, 4, 5, 6\}$$

Planning horizon (1: Jan, 2: Feb, 3: Mar, 4: Apr, 5: May, 6: Jun)

Demand Parameters

$$D_1 = 1600, D_2 = 3000, D_3 = 3200, D_4 = 3800, D_5 = 2200, D_6 = 2200$$

- Highly seasonal demand
- Capacity determined mainly by workforce size (i.e., flexible workforce)
- Subcontracting, inventories, and backlogs are allowed
- Minimizing cost is the same as maximizing profit (Why?)

Demand forecast	
Month	Demand
Jan	1600
Feb	3000
Mar	3200
Apr	3800
May	2200
Jun	2200
Total	16000



Refer to provided xlsx file

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Input data for Red Tomato Tools

Inputs	
Materials cost (\$/unit)	10
Inventory holding cost (\$/unit/month)	2
Stockout cost (\$/unit/month)	5
Hiring cost (\$/worker)	300
Layoff cost (\$/worker)	500
Labor hours per unit	4
Regular time cost (\$/hour)	4
Overtime cost (\$/hour)	6
Subcontract cost (\$/unit)	30
Capacity per worker (units/month)	40
Initial workforce W_0	80
Initial inventory I_0	1000
Minimum ending inventory (end of Jun)	500
Max overtime hours per worker per month	10

Cost Parameters

$c_m = 10$: Material cost per unit (\$/unit)

$h = 2$: Inventory holding cost per unit per month (\$/unit/month)

$b = 5$: Stockout or backlog cost per unit per month (\$/unit/month)

$c_H = 300$: Hiring cost per worker (\$/worker)

$c_L = 500$: Layoff cost per worker (\$/worker)

$c_r = 640$: Regular time labor cost per worker per month

(40 units of capacity per month x 4 hours per unit x 4 per hour = \$640/worker)

$c_o = 24$: Overtime labor cost per unit (4 hours per unit x \$6 per hour = \$24/unit)

$c_s = 30$: Subcontracting cost per unit (\$/unit)

$m = 40$: Capacity per worker (unit/month)

Capacity and Initial States

$W_0 = 80$: Initial workforce (workers)

$I_0 = 1000$: Initial inventory level (units)

$I_6 \geq 500$: Minimum required ending inventory at the end of month 6 (June)

$O_{\max} = 10$: Maximum overtime hours per worker per month (hours)



Refer to provided xlsx file

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Decision variables

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Decision variables

Decision variables		
# of Workforce in period t	W_t	$640 \cdot W_t$
Overtime hours in period t	O_t	$6 \cdot O_t$
# of newly hired in period t	H_t	$300 \cdot H_t$
# of layoff in period t	L_t	$500 \cdot L_t$
# of Inventories in period t	I_t	$2 \cdot I_t$
# of stocking out in period t	S_t	$5 \cdot S_t$
# of production units in period t	P_t	$10 \cdot P_t$
# of subcontract units in period t	C_t	$30 \cdot C_t$



Refer to provided excel file

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Linear programming model

$$\text{Minimize } \sum_{t=1}^T (c_m P_t + c_r W_t + c_o O_t + c_s C_t + h I_t + b S_t + c_H H_t + c_L L_t) \quad \text{Minimization of the total cost}$$

$$I_{t-1} - S_{t-1} + P_t = D_t + I_t - S_t \quad \forall t \quad \text{Inventory Balance Constraint}$$

$$W_t = W_{t-1} + H_t - L_t \quad \forall t \quad \text{Workforce Balance Constraint}$$

$$P_t \leq m \cdot W_t + \frac{O_t}{k} \quad \forall t \quad \text{Production Capacity Constraint}$$

$$O_t \leq O_{max} \cdot W_t \quad \forall t$$

$$I_T \geq I_{min} \quad \text{Ending Inventory Requirement}$$

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Linear programming model

$$\text{Minimize } \sum_{t=1}^T (4800W_t + 2000H_t + 2500L_t + 20I_t + 80S_t)$$

Minimization of the total cost

$$I_{t-1} - S_{t-1} + 150W_t = D_t + I_t - S_t \quad \forall t$$
$$I_0 = 500, S_0 = 0$$

Inventory Balance Constraint

$$W_t = W_{t-1} + H_t - L_t \quad \forall t$$
$$W_0 = 0$$

Workforce Balance Constraint

$$P_t \leq 150W_t \quad \forall t$$

Production Capacity Constraint